Improving Vision Transformer's Time Perfomace via Linear Attention and Token Pruning

Chi-Van Vong and Minh-Quan Pham Nguyen

University of Science, Vietnam National University Ho Chi Minh City, Vietnam
{vcvan2424,pnmquan2414}@apcs.fitus.edu.vn

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1 Introduction

In recent years, **Transformer** [15] has become a powerful foundational architecture for many models. **Vision Transformers** [2] one of the leading architectures in Computer Vision, have made significant breakthroughs in object detection, image classification, and semantic segmentation. However, softmax attention, the core mechanism of Transformer, has a high computational complexity of $O(n^2)$. To address the issue, many works attempted to group vision tokens. Although this method reduces the complexity, Transformer loses its ability to perceive information globally. Other works replace softmax attention with **linear attention** which help reduce the quadratic complexity. However, linear attention inability to model complex information yields unsatisfactory results, due to its low-rank nature and magnitude negligence. Many works have attempted to improve linear attention by tackling its low-rank nature [3]. Linear attention replaces softmax with kernal feature maps which causes the loss of sharpness. This mean that it can not be as selective as softmax attention which can place most of its attention weights on a few relevant tokens. To address this, we apply **token pruning** [12,9] which help refine the input for linear attention. With this approach, we aim to speed up the training and inference time of the model without sacrificing its performance.

2 Related Work

2.1 Vision Transformer

Transformer [15] model is initially created for NLP tasks and achieved remarkable success. However, they did not have much impact on computer vision until Alexey Dosovitskiy et al. [12] introduced the Vision Transformer (ViT), which splits the image into many small patches as tokens feed to the model. Although this approach demonstrated strong performance on many Computer Vision tasks, its quadratic complexity due to self-attention making it computationally expensive to process high resolution images. To address this, Microsoft

Research proposed Swin Transformer [8], which restricts attention to local windows, inspiring subsequent window-based variants. Other methods reduce attention cost by compressing keys or values using linear projections, strided convolutions, pooling, or patch clustering. Additionally, some variants explore alternative designs such as channel attention and softmax-free attention mechanisms.

2.2 Linear Attention

Linear Attention [14] fixes the quadratic complexity of Softmax attention by using kernel functions to approximate the $Sim()$ operator, reducing the complexity to linear. Given an input sequence of N tokens, standard self-attention computes

$$A = \text{Softmax}(QK^T)$$

where $Q, K, V \in \mathbb{R}^{N \times d}$ are the query, key, and value matrices and d is the feature dimension.

To reduce the quadratic complexity of softmax attention, linear attention replaces the softmax operation with a feature map $\phi(\cdot)$:

$$A = \phi(Q)\phi(K)^T$$

where $\phi(\cdot)$ is a kernel feature mapping applied element-wise. Due to matrix rank constraints,

$$\text{rank}(A) \leq d.$$

When the number of tokens N is significantly larger than d, the attention mechanism must compress the interactions among many tokens into a limited representation. This compression can reduce the model's ability to capture fine-grained relationships among tokens, especially in high-resolution vision tasks. Thus its performance is still largely behind Softmax. Many recent works try to tackle this issue [1,5,6,7,10,14]. FLattenTransformer [5] introduces a focused function to enhance the ability of linear attention to concentrate on relevant information, while MLLA [6] establishes a link between linear attention and the Mamba model [4]. More recently, Qihang Fan et al. [3] addresses a fundamental limitation of linear attention by identifying the low-rank nature of its feature representations as a major cause of performance degradation. To overcome this, the paper proposes Rank-Augmented Linear Attention, which enriches both the key-value buffer and output features to raise their effective rank. Many other approaches either rely on kernel-based approximations of the Softmax function or integrate depthwise convolution to improve feature diversity.

2.3 Token Pruning

ViT suffers from high computational cost propotional to the number of input tokens, hence many attempts have been made to reduce the redundant tokens

while still keeping high performance. Traditional dynamic methods like DynamicViT [12] and TokenLearner [13] adaptively drop or condense tokens. Adaptive pruning technique such as AdaViT [11] use attention metrics or learned masks to select important tokens. Recent methods introduce the use of reinforcement learning to learn the pruning policies. Specifically, Chenglong Lu et al. [9] formulates token pruning as a sequential decision making problem with multi-agent RL, where each of the visual token is consider an agent.

3 Method

We propose a reinforcement-learning (RL) based token routing framework with a shared memory bank to improve the effectiveness of linear attention when applying to a ViT for high resolution images.

3.1 Overall Architecture

Our study augments a standard ViT with a reinforcement-learning based token routing module and linear attention mechanism. Given an input image, the ViT patch embedding first converts the image into a sequence of N patch tokens. Instead of processing all tokens through every transformer layer, we introduce a lightweight policy network π_θ that dynamically routes tokens at selected stage transitions. The policy assigns each token an action, that is: *keep*, *remove*, *store*, or *retrieve*, allowing the model to filter redundant tokens and preserve informative ones. Tokens assigned the *keep* or *retrieve* actions are packed into a shortened sequence of length N_{active} and passed to the subsequent linear attention blocks. The attention output is computed as

$$O = \phi(Q_{\text{active}})\big(\phi(K_{\text{active}})^T V_{\text{active}}\big).$$

We aim to make $N_{\text{active}} \ll N$, the effective computational cost is reduced while maintaining the model's ability to focus on important tokens. Meanwhile, contextual information from discarded or stored tokens can be preserved through the shared memory mechanism, allowing global context to bypass the linear attention bottleneck.

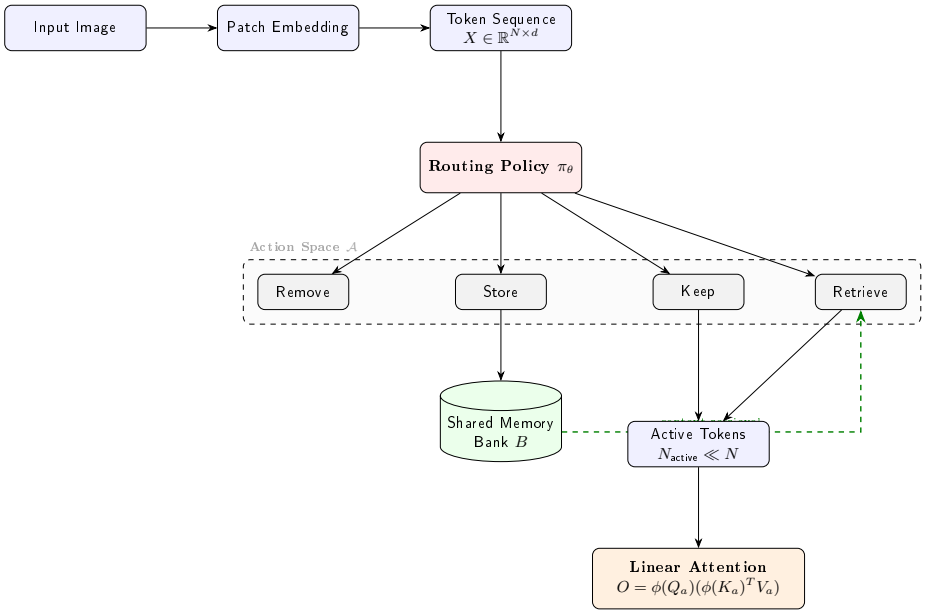

Fig. 1. Overall Method

3.2 RL-Based Token Routing

To tackle the limitations of linear attention, we introduce a lightweight RL policy network that dynamically routes tokens during inference. Given a token representation $x_i \in \mathbb{R}^d$, the policy network π_θ computes an action distribution

$$\pi_\theta(a_i \mid x_i) = \text{Softmax}(W_2 \sigma(W_1 x_i)),$$

where W_1 and W_2 are learnable parameters and $\sigma(\cdot)$ denotes a nonlinear activation function. The policy selects an action

$$a_i \in \{\text{keep}, \text{remove}, \text{store}, \text{retrieve}\}.$$

Each action modifies the token flow as follows:

- **Keep**: the token remains in the active sequence and participates in linear attention.
- **Remove**: the token is discarded to reduce sequence length.
- **Store**: the token is compressed into a shared memory bank.
- **Retrieve**: the token receives contextual information from the memory bank.

Let $\mathcal{A}$ denote the set of tokens selected by the policy to remain active. Linear attention is then computed only over this subset, which reduces the effective token count $N_{\text{active}} = |\mathcal{A}|$. The policy is trained with reinforcement learning to

maximize a reward that balances model performance and computational efficiency.

3.3 Shared Memory Bank

Tokens assigned to the *store* action are aggregated into a shared memory representation:

$$B = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} x_j,$$

where $\mathcal{S}$ denotes the set of stored tokens.

Tokens requesting contextual information retrieve it through a residual update

$$\hat{x}_i = x_i + W_r B,$$

where W_r is a learnable projection matrix.

This memory pathway allows global contextual information to bypass the linear attention bottleneck while keeping the computational cost low.

3.4 Training Objective

The routing policy is trained using reinforcement learning. The reward function balances two objectives: preserving downstream task performance, minimizing the number of active tokens. The reward is defined as

$$R = \alpha \, \mathrm{Acc} - \beta \frac{N_{\text{active}}}{N},$$

where Acc represents task performance, and α, β control the trade-off between accuracy and efficiency.

4 Experiments

- We conduct experients on multiple vision task, namely, object detection and image classification.
- We evaluate model's training and inference efficiency to understand the trade-off.
- We conduct ablation studies to validate the importance of each component.

4.1 Image Classification

Settings

4.2 Object Detection

4.3 Training and Inference Speed

5 Results

6 Discussion

7 Conclusion

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